

Jira Platform

Architecture Reference

Per-Service API Map
Issue Create & Workflow Transition Sequences
Gateway Route Gap Analysis

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1. Platform Overview

The Jira Platform is a microservices system: React SPA (port 3000) -> Spring Cloud Gateway (8080) -> 14 domain services (8081-8094) -> PostgreSQL (per-service databases). JWT authentication is enforced at the gateway; identity is forwarded via X-User-Id, X-Username, and X-User-Roles headers.

1.1 Service Ports

Service	Port	DB
jira-gateway	8080	-
jira-auth-service	8081	auth_db
jira-user-service	8082	user_db
jira-project-service	8083	project_db
jira-issue-service	8084	issue_db
jira-workflow-service	8085	workflow_db
jira-comment-service	8086	comment_db
jira-notification-service	8087	notification_db
jira-search-service	8088	search_db
jira-audit-service	8089	audit_db
jira-attachment-service	8090	attachment_db
jira-sprint-service	8091	sprint_db
jira-plan-service	8092	plan_db
jira-admin-service	8093	admin DB
jira-migration-service	8094	migration_db
jira-frontend	3000	-

2. Current Gateway Routes (local profile)

Defined in jira-gateway/src/main/resources/application-local.yml

Path Predicate	Target	Filters
/api/auth/**	:8081	StripPrefix=1
/api/users/**	:8082	-
/api/projects/**	:8083	-
/api/issues/**	:8084	-
/api/components/**	:8084	-
/api/versions/**	:8084	-
/api/workflows/**	:8085	-
/api/comments/**	:8086	StripPrefix=1
/api/notifications/**	:8087	-
/api/search/**	:8088	-
/api/audit/**	:8089	-
/api/attachments/**	:8090	-
/api/sprints/**	:8091	-
/api/plans/**	:8092	-

3. Per-Service API Map

3.1 Auth Service (:8081)

Base: /auth (gateway: /api/auth -> StripPrefix)

Method	Path	Description
POST	/auth/register	Register user
POST	/auth/login	Login, returns JWT
POST	/auth/refresh	Refresh access token
GET	/auth/me	Current user profile

3.2 User Service (:8082)

Base: /api/users

Method	Path	Description
GET	/api/users	List/search users
GET	/api/users/{id}	Get user
POST	/api/users	Create user
PUT	/api/users/{id}	Update user
DELETE	/api/users/{id}	Delete user

3.3 Project Service (:8083)

Base: /api/projects

Method	Path	Description
GET/POST	/api/projects	List / create projects
GET/PUT/DELETE	/api/projects/{id}	CRUD project
POST	/api/projects/{id}/archive	Archive project
GET	/api/projects/{id}/members	Project members
POST	/api/projects/{id}/members	Add member

3.4 Issue Service (:8084)

Base: /api/issues (+ components, versions)

Method	Path	Description
POST	/api/issues	Create issue (requires X-User-Id)
GET	/api/issues	Search/list with filters
GET	/api/issues/{id}	Get issue
PUT	/api/issues/{id}	Update issue
PATCH	/api/issues/{id}/status	Status transition (validates workflow)
DELETE	/api/issues/{id}	Delete issue
GET	/api/issues/search	JQL search (in-issue service)
*	/api/issues/{id}/worklogs	Worklog CRUD
*	/api/issues/{id}/labels	Label CRUD
*	/api/issues/{id}/links	Issue links

*	/api/issues/{id}/history	Change history
*	/api/components	Component CRUD
*	/api/versions	Version CRUD

3.5 Workflow Service (:8085)

Base: /api/workflows + /api/admin/workflows

Method	Path	Description
POST	/api/workflows	Create workflow
GET	/api/workflows/project/{projectId}	Project workflows
GET	/api/workflows/project/{pid}/validate-transition	Validate from/to status
POST	/api/workflows/issues/{issueId}/execute	Execute transition
GET	/api/workflows/{wfid}/allowed-transitions	Allowed transitions
*	/api/admin/workflows/**	Admin: schemes, screens, versions, audit

3.6 Comment Service (:8086)

Gateway strips /api prefix

Method	Path	Description
POST	/comments	Create (frontend: /api/comments)
GET	/comments/issue/{issueId}	List by issue
PUT/DELETE	/comments/{id}	Update/delete

3.7 Notification Service (:8087)

Base: /api/notifications

Method	Path	Description
GET	/api/notifications	List notifications
PUT	/api/notifications/{id}/read	Mark read

3.8 Search Service (:8088)

Base: /api/jql (NOT /api/search in controller)

Method	Path	Description
POST	/api/jql/search	JQL search execution
GET	/api/jql/validate	Validate JQL syntax

3.9 Audit Service (:8089)

Base: /api/audit

Method	Path	Description
GET	/api/audit/logs	Search audit logs
GET	/api/audit/logs/{type}/{id}	Entity audit trail
POST	/api/audit/logs	Create audit entry

3.10 Attachment Service (:8090)

Base: /api/attachments

Method	Path	Description
POST	/api/attachments	Upload (multipart)
GET	/api/attachments/issue/{issueId}	List by issue
GET	/api/attachments/{id}/download	Download file
DELETE	/api/attachments/{id}	Delete

3.11 Sprint Service (:8091)

Agile boards, sprints, dashboards

Method	Path	Description
*	/api/sprints/**	Sprint CRUD, start/complete
*	/api/boards/**	Agile board config & issues
*	/api/dashboards/**	Dashboard & gadgets
*	/api/filters/**	Saved JQL filters
*	/api/bulk-operations/**	Bulk issue operations

3.12 Plan Service (:8092)

Programs, plans, Advanced Roadmaps

Method	Path	Description
*	/api/plans/**	Plans, backlog, teams, releases
*	/api/plans/programs/**	Program management
*	/api/plans/boards/**	Plan board config & sprints
*	/api/plans/sprints/**	Plan-scoped sprints
*	/api/plans/working-days/**	Working days & holidays

3.13 Admin Service (:8093)

Base: /api/admin

Method	Path	Description
*	/api/admin/settings	System settings
*	/api/admin/users/**	User administration
*	/api/admin/issues/**	Issue types, priorities, schemes
*	/api/admin/projects/**	Project admin
*	/api/admin/datacenter/**	DC cluster settings
*	/api/admin/audit/**	Admin audit views

3.14 Migration Service (:8094)

Base: /api/migration

Method	Path	Description
POST	/api/migration/import/csv	CSV import
POST	/api/migration/import/jira-dc	Jira DC import

GET	/api/migration/jobs/{id}	Job status
*	/api/migration/wizard/**	Import wizard steps
*	/api/sse/**	Server-sent events for progress

4. Sequence: Issue Create

Flow when user creates an issue from the React UI via `issueApi.create()`.

```
Actor: User (Browser)
|
| POST /api/issues { projectId, title, issueTypeId, ... }
| Authorization: Bearer <JWT>
v
jira-gateway (:8080)
|-- Validate JWT, extract X-User-Id, X-Username, X-User-Roles
|-- Route to issue-service (:8084)
v
IssueController.createIssue(request, X-User-Id)
v
IssueService.createIssue()
|-- GET project-service: resolve project key for issue key (e.g. PROJ-42)
|   RestTemplate -> http://localhost:8083/api/projects/{projectId}
|-- Load IssueType, IssueStatus (default Open), IssuePriority from issue_db
|-- generateIssueKey(projectKey)
|-- issueRepository.save(issue)
v
201 Created IssueResponse { id, issueKey, status, ... }

Optional downstream (not always synchronous):
- search-service index (if wired)
- audit-service log entry
- notification-service (assignee notify)
```


5. Sequence: Workflow Status Transition

Two paths exist: (A) simple status patch on issue service, (B) full workflow execute.

Path A - PATCH `/api/issues/{id}/status` (used by `issueApi.transitionStatus`)

```
IssueService.updateIssueStatus():  
  1. Load issue, get current status  
  2. validateTransition(projectId, fromStatus, toStatus)  
      GET workflow-service:  
        /api/workflows/project/{projectId}/validate-transition  
        ?fromStatus=X&toStatus=Y  
  3. If invalid -> 400 InvalidTransitionException  
  4. Update issue.status, save, return IssueResponse
```

Browse

Path B - POST `/api/workflows/issues/{issueId}/execute` (workflow engine)

```
WorkflowService.executeTransition():  
  1. Load transition, workflow, issue (may call issue-service)  
  2. Evaluate conditions (role, field value, etc.)  
  3. Run validators  
  4. Apply post-functions (update fields, assign, etc.)  
  5. Update issue status to transition target  
  6. Return TransitionExecutionResponse
```

Browse

Note: Frontend `issueApi` uses Path A. Admin workflow designer uses `/api/admin/workflows` on `workflow-service`. `WorkflowsPage.tsx` incorrectly calls port 8082 (user service).

6. Gateway Route Gap Analysis

Comparison of frontend API calls (jira-frontend/src/api/* and hooks) against gateway routes in application-local.yml.
Status: MISSING = no gateway route.

Priority	Frontend Path	Service Port	Used By
CRITICAL	/api/admin/**	8093	adminApi, useAdminApi, admin pages
CRITICAL	/api/admin/workflows/**	8085	WorkflowsPage (also wrong port 8082)
CRITICAL	/api/migration/**	8094	migrationApi, MigrationPage
CRITICAL	/api/jql/**	8088	serviceApi.searchApi.jqlSearch
HIGH	/api/boards/**	8091	boardApi, KanbanBoard, BoardsPage
HIGH	/api/dashboards/**	8091	dashboardApi, DashboardPage
HIGH	/api/filters/**	8091	filterApi, SavedFilters
HIGH	/api/bulk-operations/**	8091	bulkApi
MEDIUM	/api/custom-fields/**	8084?	issueApi custom fields - verify backend
MEDIUM	/api/resolutions/**	8084?	issueApi resolutions - verify backend
MEDIUM	/search (no /api prefix)	8088	serviceApi.searchApi.search
LOW	/api/ws/**	8094	migration WebSocket
LOW	/api/sse/**	8094	migration SSE progress
CONFIG	/api/comments/**	8086	StripPrefix=1 may break path /api/comments/issue/{
CONFIG	/api/search/**	8088	Routed but controller uses /api/jql not /api/searc

6.1 Recommended Gateway Routes to Add

```
- id: admin-service
  uri: http://localhost:8093
  predicates: [Path=/api/admin/**]
  # Exclude /api/admin/workflows if routing to workflow-service

- id: workflow-admin-service
  uri: http://localhost:8085
  predicates: [Path=/api/admin/workflows/**]

- id: migration-service
  uri: http://localhost:8094
  predicates: [Path=/api/migration/**, /api/sse/**, /api/ws/**]

- id: jql-service
  uri: http://localhost:8088
  predicates: [Path=/api/jql/**]

- id: boards-service
  uri: http://localhost:8091
  predicates:
    - Path=/api/boards/**, /api/dashboards/**, /api/filters/**, /api/bulk-operations/**
```

6.2 Frontend Fixes Required

- WorkflowsPage.tsx: change API_BASE from localhost:8082 to gateway /api/admin/workflows

- ser

6.3 Comment Service StripPrefix Issue

Gateway strips first path segment (/api) so /api/comments becomes /comments at comment-service. CommentController uses /comments paths - this works for POST /api/comments. GET /api/comments/issue/{id} becomes /comments/issue/{id} - verify controller mapping matches.

7. Inter-Service Communication

Caller	Callee	Purpose
issue-service	project-service	Project key for issue key generation
issue-service	workflow-service	Validate status transitions
search-service	issue-service	JQL result hydration
notification-service	user-service	Recipient lookup
migration-service	all services	Import orchestration via RestTemplate clients
sprint/board-service	issue-service	Board issue data
workflow-service	issue-service	Transition execution updates

8. Frontend Feature -> API Mapping

Feature	Primary API Modules	Gateway Prefix
Auth	authApi	/api/auth/**
Projects	projectApi, issueApi.projectApi	/api/projects/**
Issues	issueApi, labelApi, worklogApi, changeHistoryApi	/api/issues/**
Boards/Kanban	boardApi	/api/boards/** MISSING
Sprints	sprintApi	/api/sprints/**
Plans	planApi, useSprint, useLexoRank	/api/plans/**
Search	serviceApi.searchApi	/api/jql/** MISSING
Admin	adminApi, useAdminApi	/api/admin/** MISSING
Migration	serviceApi.migrationApi	/api/migration/** MISSING
Workflows UI	WorkflowsPage fetch	BROKEN (wrong host)